

RESEARCH ARTICLE

Effects of caffeine and sex on muscle performance and delayed-onset muscle soreness after exercise-induced muscle damage: a double-blind randomized trial

Hou-Yu Chen,¹ Yung-Chih Chen,^{2,4} Kang Tung,² Hsiao-Han Chao,³ and Ho-Seng Wang²

¹Education Center for Humanities and Social Sciences, National Yang Ming University, Taipei, Taiwan; ²Department of Physical Education, National Taiwan Normal University, Taipei, Taiwan; ³Department of Athletics, National Taiwan University, Taipei, Taiwan; and ⁴Department for Health, University of Bath, Bath, United Kingdom

Submitted 17 December 2018; accepted in final form 18 June 2019

Chen HY, Chen YC, Tung K, Chao HH, Wang HS. Effects of caffeine and sex on muscle performance and delayed-onset muscle soreness after exercise-induced muscle damage: a double-blind randomized trial. *J Appl Physiol* 127: 798–805, 2019. First published June 20, 2019; doi:10.1152/jappphysiol.01108.2018.—The present study aims to investigate effects of caffeine ingestion and sex difference on muscle performance, delayed-onset muscle soreness (DOMS), and various biomarkers under exercise-induced muscle damage (EIMD). Twenty (10 male and 10 female) healthy elite college athletes were recruited. Participants ingested either caffeine (6 mg/kg) or a placebo in a randomized, double-blind, and counterbalanced fashion at 24 and 48 h following EIMD. Muscle performance, DOMS, and blood samples were taken an hour before and an hour after supplementation. Caffeine ingestion restored impaired maximal voluntary isometric contractions (MVIC: 10.2%; MVIC_{post}: 7.2%, both $P < 0.05$) during EIMD across both sexes. Following caffeine ingestion during MVIC, while affected by EIMD, an interaction was found in DOMS and serum K⁺ (both $P < 0.05$), with males showing greater attenuation (21.5 and 16.9%, respectively) compared with females (4.6 and 1.3%, respectively). DOMS demonstrated an inverse correlation with MVIC after caffeine ingestion both overall and among male athletes ($r = -0.34$ and -0.54 , respectively; $P < 0.05$) but not among female athletes ($r = -0.11$; $P > 0.05$) under EIMD. In addition, caffeine ingestion increased postexercise serum glucose and lactate concentrations across both sexes (both $P < 0.05$). This is the first study to show that male athletes, compared with female athletes, experience a greater reduction in DOMS during enhanced MVIC when caffeine was consumed, suggesting men might receive greater ergogenic effects from caffeine when affected by EIMD. Furthermore, caffeine ingestion was able to restore impaired muscle power among elite collegiate athletes across both sexes.

NEW & NOTEWORTHY Exercise-induced muscle damage (EIMD) reduces anaerobic/aerobic performance and increases delayed-onset muscle soreness (DOMS) during exercise. We show that acute caffeine supplementation at a dosage of 6 mg/kg seems to facilitate recovery of anaerobic muscle power and attenuate DOMS after EIMD across both sexes. Furthermore, male athletes, compared with female athletes, when caffeine was prescribed, experience a greater reduction in DOMS with better restoration of impaired maximal voluntary isometric contractions. This suggests that male athletes might benefit from the ergogenic effect of acute caffeine supplementation after the onset of EIMD.

adenosine receptor; creatine kinase; DOMS; ergogenic; MVIC

INTRODUCTION

Delayed-onset muscle soreness (DOMS), a symptom that involves a perception of severe soreness and discomfort, refers to exercise-induced muscle damage (EIMD) that occurs 24–72 h after unaccustomed and/or high-intensity eccentric contractions (12). The symptom can be attributed to local inflammation induced by morphological damage to myofibers; it is usually assessed by examining fluctuations in the serum level of creatine kinase (CK) (11). Osmotic swelling and vacuole formation within the t-tubules may also contribute to DOMS (29). It has been suggested that exercise under EIMD conditions may transiently lead to a decrease in muscle functionality such as reduced muscle strength and/or explosive power (4, 7, 31), partly due to the activation of Ca²⁺-dependent mechanisms, causing proteolysis (i.e., less Ca²⁺ release from the sarcoplasmic reticulum) (34). Unavoidably, most athletes are highly likely to experience DOMS when they take part in consecutive days of competition. Studies have reported that a range of treatments, such as medication (non-steroidal anti-inflammatory drugs), physical therapy (massage, stretching), and other approaches (antioxidant, ultrasound) have only limited effects in terms of attenuating DOMS and reducing the deterioration in muscle function during exercise when EIMD is present (11, 12). Therefore, alternative strategies are needed to reduce DOMS and help recovery from EIMD; this would help to improve athletic performance in these circumstances.

In the last few decades, caffeine supplementation has been used as a legal ergogenic aid that is able to enhance endurance performance (5, 17). In addition, caffeine has been shown to improve muscle strength (22, 23, 45) and explosive power (23) via 1) an enhanced Ca²⁺ efflux from the sarcoplasmic reticulum (28), 2) a direct Ca²⁺-sensitizing effect on skeletal muscle (43), and 3) an increase in Na⁺-K⁺-ATPase activity resulting from decreased serum levels of K⁺ (39). Direct ergogenic effects can also be found within the skeletal muscle after caffeine supplementation (14, 40) [see review for more details (41, 43)]. Moreover, caffeine is similar in structure to adenosine, and it has been proposed that it is able to inhibit the effect of adenosine on the central nervous system (CNS) by blocking

Address for reprint requests and other correspondence: H.-S. Wang, National Taiwan Normal Univ., PO Box 97-87, Taipei City 11699, Taiwan (e-mail: hoseng@ntnu.edu.tw).

the perception of pain that is propagated from the peripheral nerves to the CNS via adenosine receptors. This attenuating effect of caffeine has been supported by clinical studies involving chronic headache, migraine, and postoperative pain (38). Recent studies further demonstrated that caffeine ingestion is able to, first, decrease pain perception in quadriceps during moderate-intensity and high-intensity cycling (17–19), second, to attenuate DOMS during exercise under the EIMD condition (25, 26, 31), third, to facilitate recovery of performance after carrying out knee extension maximal voluntary isometric contractions (MVIC) (31), and, finally, to enhance the total number of repetitions when bicep curls are performed (25).

It is important to note that the beneficial effects of caffeine on performance during knee extension MVIC, which acts as an indicator of muscle anaerobic power, was inconsistent under non-EIMD condition when trained athletes and untrained individuals were compared (2, 6, 30, 35, 42). By way of contrast, it has been shown that caffeine intake under non-EIMD conditions has an ergogenic effect on submaximal voluntary isometric contractions fatigue (T_{lim}), which is an indicator of muscle anaerobic endurance, and that this occurs regardless training status (6, 30, 32, 35). However, only one study has pinpointed the effect of caffeine on recovery of performance after MVIC under EIMD conditions among nonathletes (31) and that no previous studies have examined the effect of caffeine on T_{lim} performance under EIMD conditions. Thus additional work is warranted that explores both MVIC and T_{lim} to gain a more comprehensive view of muscle anaerobic power and muscle endurance under EIMD conditions, in particular focusing on athletes who train regularly.

Another issue also warrants further investigation, and this is the effect of sex and the relationship between caffeine supplementation and DOMS under EIMD conditions. Studies have reported that sex may affect an individual's perception of pain, expression of pain, and tolerance of pain via differences in psychosocial experience (33). Paradoxically, sex difference in DOMS or muscle pain perception was diminished in the exercise situation with(out) muscle damage (24, 36). Given that previous investigations have recruited either only female individuals (26, 31) or only male individuals (19), the moderating role of sex in the relationship between caffeine supplementation and DOMS under EIMD conditions remains unclear (21).

Taken together, the purposes of current study was to investigate the effects of caffeine and sex on muscle anaerobic performances (MVIC and T_{lim}), DOMS, and circulating biomarkers under EIMD conditions. We hypothesized that caffeine supplementation will improve anaerobic muscle performance and attenuate DOMS under EIMD conditions, and that these effects of caffeine may be moderated by sex.

MATERIALS AND METHODS

Participants

Twenty (10 male and 10 female) division-one college athletes (sports: tennis, basketball, English football) were recruited into the current study. All the female participants had normal menstrual cycles. The inclusion criteria were as follows: 1) no regular caffeine consumption (<200 mg/wk) by self-reporting in response to a questionnaire; 2) nonsmokers; 3) no regular taking of any medicine or central nervous stimulator that could potential affect the results of the study; and 4) no allergic symptoms when exposed to caffeine. Before taking part, all participants signed an informed consent form approved by the Taipei Medical University's institutional review board in accordance with the Declaration of Helsinki. Table 1 shows the demographics of participants.

Experimental Procedures

Experimental design. Four consecutive testing sessions were conducted on separate days, and the participants had refrained from caffeine 7 days before the first testing session. At the first session, measurements assessing the participants' anthropometric data, baseline data regarding muscle performance, DOMS, and their baseline serum biomarkers were taken, which was followed by testing of maximum oxygen consumption ($\dot{V}O_{2max}$). During the second session, which was arranged 2 days after the first session, the participants performed the EIMD-inducing paradigm, which consisted of a 30-min bout of downhill running at a declination of 15% and an intensity of 70% $\dot{V}O_{2max}$, the aim being to induce muscle damage within the quadriceps. Measurements of the dependent variables, namely muscle performance, DOMS and serum biomarkers, were taken at the third session (24 h after EIMD was induced), which was followed by ingestion of a supplement containing either caffeine or a placebo. Half of the participants (5 male and 5 female) were administered a caffeine supplement, while the others received a placebo supplement; this was carried out in a double-blinded and counterbalanced manner. One hour after the supplements, measurement of the dependent variables was taken once again. At the fourth session (48 h after EIMD was induced), the participants who had received a caffeine supplement during the prior session were given the placebo, and vice versa, and then the rest of the procedures were carried out in the same manner as during the third session. Figure 1A summarizes the experimental procedures.

Experimental control. CIRCADIAN RHYTHMS. To avoid circadian rhythm affecting the blood biomarkers of the participants, all test trials were conducted between 8 and 10 AM.

DIETS. Participants were required to have undertaken an at least 8- to 10-h overnight fast before the tests were conducted. In addition, other than for testing session 2, a standardized breakfast (353 kcal; carbohydrate 67.6%, fat 23.1%, and protein 9.3%) was provided 30 min before the exercise tests, and this was followed by the taking of a baseline blood sample.

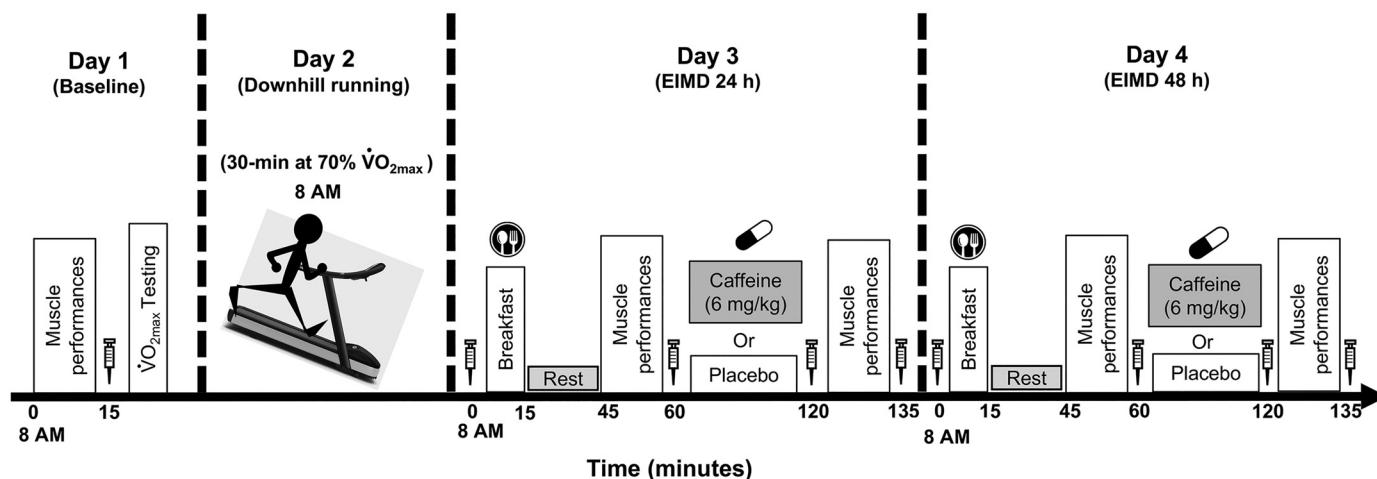
MENSTRUATION. To avoid the confounding effect of the participants' menstrual cycle on caffeine metabolism, all female participants were instructed to participate in the testing during their early follicular phase. They were also instructed to avoid using any form of contra-

Table 1. Descriptive data of participants

	Age, yr	Height, cm	Mass, kg	BMI, kg/m ²	$\dot{V}O_{2max}$, ml·kg ⁻¹ ·min ⁻¹
Male (<i>n</i> = 10)	21.1 ± 2.1	180.2 ± 4.9*	73.1 ± 7.5*	22.4 ± 1.5	51.1 ± 6.5
Female (<i>n</i> = 10)	20.4 ± 1.2	166.4 ± 5.4	58.4 ± 5.1	21.1 ± 1.6	46.4 ± 2.0

Values are means ± SD. BMI, body mass index; $\dot{V}O_{2max}$, maximum oxygen consumption. *Male greater than female (*P* < 0.05).

A Experimental procedures



EIMD: exercise-induced muscle damage

: supplements

: diets

B Muscle performances protocol

Warm up	Recovery 3~5 min	MVIC × 3	DOMS	Rest 5 min	T_{lim}	Rest 20 sec	MVIC _{post} × 1
---------	---------------------	----------	------	---------------	-----------	----------------	--------------------------

DOMS: perceived muscle soreness

MVIC: maximal voluntary isometric contractions

T_{lim} : submaximal voluntary isometric fatigue protocol

MVIC_{post}: after T_{lim} and rest 20 seconds, do MVIC again.

Fig. 1. Experimental design.

ception that would affect their hormone responses during the experiments.

Supplements. A capsule of 500 ml of water mixed with 6 mg/kg (25) caffeine (Sigma-Aldrich, Sydney, Australia) or a placebo (diet flour, Sun Right, Taiwan) was administered to the participants.

$\dot{V}O_{2max}$ test. An incremental continuous treadmill (h/p/cosmos treadmill (Mercury, h/p/cosmos, Nussdorf, Traunstein, DE) test until the point of volitional fatigue was used to determine $\dot{V}O_{2max}$. An automatic gas analysis system ($V_{max}29c$, SensorMedics, Yorba Linda, CA) was employed throughout the test. Participants warmed up at a speed of 4 mi/h on a gradient of 0% for 5 min, and then the speed was increased by 1 mi/h every 3 min until the speed reached 10 mi/h; at the same time the gradient was increased by 2% every 3 min until fatigue (9).

Downhill running. The protocol was performed 2 days after the $\dot{V}O_{2max}$ testing. All participants performed a 30-min downhill run on the treadmill to induce muscle damage (9). The velocity was adjusted to correspond to the predetermined 70% $\dot{V}O_{2max}$ target of each participant, which had been obtained by linear regression analysis. The averaged velocities were 169 ± 7 m/min and 149 ± 8 m/min for male and female individuals, respectively. The gradient of the treadmill was set at -15% throughout.

Measurement of muscle performance. Muscle performance of the dominant knee extensor/flexor was measured by a dynamometer (Biodex Medical Systems). The measurements were taken at the following five time points: baseline, and before/after supplementation at 24 and 48 h following EIMD. Participants were seated on the dynamometer so that the hip was flexed at 85° with the thigh in a

horizontal position. Next, participants warmed up by performing three sets of 10 consecutive isokinetic knee extensions at 240, 120, and 30°/s (knee range of motion of these contractions was $0-90^\circ$). After this, the participants performed three MVIC tests at a knee joint angle of 70° with a 3- to 5-min recovery. Each test lasted for 5 s, with 1-min intervals. The peak power obtained for the three tests was identified as the MVIC. T_{lim} was calculated 5 min after the MVIC test. The participants performed an individual 50% MVIC, and the T_{lim} was identified when they could no longer maintain their power output at 45% MVIC for 3 s for each individual (3). The MVIC_{post} was tested 20 s after the T_{lim} test, and during this the participants had to perform another MVIC. Fatigue index % (FI%) was calculated by the $[100\% - (MVIC_{post}/MVIC)] \times 100\%$ (30). Figure 1B represents the muscle performance testing protocol. Verbal encouragements were given to participants during the experimental process.

Blood sampling and analysis. Five-milliliter blood samples were taken from the antecubital vein at the following nine time points: baseline, and 1 h before and after supplements given at 24 and 48 h following EIMD, and 3 min after muscle performance tests (four exercise tests in total after EIMD). The blood samples were centrifuged at 3,000 rpm and then stored at -80°C for future analysis. Various serum biomarkers, namely K^+ , lactate, glucose, and CK, were analyzed using appropriate specific assays (Vitros DT60 II chemistry system, Ortho Clinical Diagnostics).

DOMS. DOMS ratings were measured using a horizontal visual scale that consisted of graduations from 0 to 100. The left end was marked "scale 0", which represented "no soreness at all," while the

right end was marked “scale 100”, which represented “extreme soreness.” Participants were instructed to mark their subjective ratings for DOMS regarding knee extension; this took place after the MVIC tests, both before supplementation and after supplementation 24 and 48 h following the induction of EIMD (31).

Statistical Analysis

Statistical analyses were performed on all dependent variables using SPSS version 22.0 for Windows. All data are presented as means $\pm$ SD. First, mixed-design 2 (sex: male, female) $\times$ 2 (time: pre-EIMD, post-EIMD) ANOVAs were applied to all dependent variables (MVIC, T_{lim} , MVIC_{post}, FI%, CK) to identify performance loss and inflammation after EIMD. Second, each individual dependent measure was examined before, at 24 h after, and at 48 h after EIMD induction under either the caffeine or placebo condition. Next, 2 (sex: male, female) $\times$ 2 (supplement: caffeine, placebo) ANCOVAs were then applied to muscle performance data, DOMS, and circulating blood biomarkers. At this point body weight and other dependent variables measured at the presupplement time point were used as covariates. Pearson correlations were used to determine the relationships between DOMS and MVIC (overall, male and female, separately) in terms of change score (%) after caffeine supplementation under EIMD. The level of statistical significance was set at $\alpha = 0.05$, and the effect sizes were calculated using Cohen's d formula.

RESULTS

Performance Loss and CK Concentrations After EIMD

Table 2 summarizes the results for muscle performance and circulating CK concentration under pre-EIMD and post-EIMD conditions. For all muscle performance results, the interaction (sex $\times$ time) and the main effect of sex were not significant ($P > 0.05$). For the main effect of time, the MVIC ($d = 0.76$), T_{lim} ($d = 0.61$), and MVIC_{post} ($d = 0.69$) performances for knee extensions were inferior under the post-EIMD condition compared with the pre-EIMD condition ($P < 0.001$). A significant interaction (sex $\times$ time) was found for CK serum levels ($P < 0.01$, $d = 0.29$). Post hoc analysis revealed that males had higher serum levels of CK compared with females ($P < 0.05$, $d = 0.55$) under the post-EIMD condition. Furthermore, when the main effect of time was examined, the serum levels of CK were higher under the post-EIMD condition compared with the pre-EIMD condition ($P < 0.001$, $d = 0.65$).

Effect of Sex and Caffeine Ingestion on Muscle Performance, DOMS, and Circulating Serum Biomarkers after EIMD

Post-EIMD measures of muscle performance, DOMS and serum biomarker concentrations after caffeine and placebo supplementation are presented in Fig. 2. For muscle perfor-

mance, neither an interaction effect (sex $\times$ supplementation) nor a main effect of sex was found ($P > 0.05$). When the main effect of supplement was examined, caffeine ingestion revealed benefits in terms of performance of $\sim 10.2\%$ for MVIC (caffeine: 3.55 ± 0.87 Nm/kg; placebo: 3.22 ± 0.81 Nm/kg; $P < 0.01$, $d = 0.48$) and 7.2% for MVIC_{post} (caffeine: 2.98 ± 0.79 Nm/kg; placebo: 2.78 ± 0.73 Nm/kg; $P < 0.05$, $d = 0.34$), both compared with the placebo.

When DOMS was investigated, the interaction (sex $\times$ supplementation) was found to be significant ($P < 0.05$, $d = 0.30$). Post hoc analysis revealed that the decrease in DOMS was greater in males (21.5%) than in females (4.6%) after caffeine supplementation ($P < 0.01$, $d = 0.61$). In terms of the main effect of supplementation, DOMS was lower after caffeine supplementation (75.25 ± 13.03) than that after placebo supplementation (86.50 ± 6.90 , $P < 0.001$, $d = 0.50$).

When the serum biomarkers were explored, the interaction (sex $\times$ supplementation) was only significant for circulating K^+ concentration ($P < 0.01$, $d = 0.40$). Post hoc analysis revealed that the decrease in serum levels of K^+ was greater in males (16.9%) than in females (1.3%; $P < 0.01$, $d = 0.54$). When the main effect of supplementation was examined, serum levels of K^+ were lower after caffeine ingestion (3.61 ± 0.49 mmol/L) than that after placebo ingestion (3.97 ± 0.74 mmol/L; $P < 0.01$, $d = 0.48$), whereas the levels of glucose (caffeine: 4.46 ± 0.56 mmol/L; placebo: 4.01 ± 0.44 mmol/L; $P < 0.01$, $d = 0.41$) and lactate (caffeine: 4.56 ± 2.03 mmol/L; placebo: 3.64 ± 1.59 mmol/L; $P < 0.001$, $d = 0.59$) were higher after caffeine supplementation than after placebo supplementation. No main effect of sex was found for circulating glucose and lactate concentrations.

Relationships Between Variables

There was an inverse correlation after caffeine supplementation for change score (%) for both sexes ($r = -0.34$; $P = 0.05$) and for male participants ($r = -0.54$; $P = 0.03$) but not for female participants ($r = -0.11$; $P = 0.37$) between the DOMS index and MVIC under EIMD.

DISCUSSION

This study investigated the impact of acute caffeine ingestion after EIMD by exploring muscle performance (anaerobic power and endurance), DOMS, and circulating biomarkers among an elite group of male and female college athletes. Our results demonstrated that caffeine supplementation at a dosage of 6 mg/kg an hour before exercise effectively restored the impaired knee extension muscle power during MVIC (10.2%)

Table 2. Effect of sex on pre-EIMD and post-EIMD muscle performance and circulating serum CK concentrations

	Male		Female		All Athletes		$\Delta\%$
	pre-EIMD	post-EIMD	pre-EIMD	post-EIMD	pre-EIMD	post-EIMD	
MVIC, Nm/kg	4.47 $\pm$ 0.57	3.53 $\pm$ 0.71	3.94 $\pm$ 0.43	3.00 $\pm$ 0.71	4.21 $\pm$ 0.56	3.26 $\pm$ 0.74 ^a	-22.6
T_{lim} , s	85.70 $\pm$ 33.23	59.40 $\pm$ 12.94	87.20 $\pm$ 19.09	54.90 $\pm$ 14.76	86.45 $\pm$ 26.39	57.15 $\pm$ 13.70 ^a	-33.9
MVIC _{post} , Nm/kg	3.75 $\pm$ 0.74	2.93 $\pm$ 0.71	3.35 $\pm$ 0.49	2.58 $\pm$ 0.78	3.55 $\pm$ 0.65	2.75 $\pm$ 0.74 ^a	-22.5
Fatigue index, %	16.72 $\pm$ 7.76	17.40 $\pm$ 6.36	15.18 $\pm$ 7.53	14.96 $\pm$ 9.87	15.95 $\pm$ 7.48	16.18 $\pm$ 8.17	0.09
CK, U/L	153.40 $\pm$ 31.28	940.70 $\pm$ 543.67 ^c	128.50 $\pm$ 41.60	412.60 $\pm$ 221.05	140.95 $\pm$ 38.04	676.65 $\pm$ 486.36 ^b	380

Values are means $\pm$ SD. MVIC, maximal voluntary isometric contractions; T_{lim} , submaximal voluntary isometric fatigue protocol; MVIC_{post}, after T_{lim} and rest 20 s, do MVIC again; Fatigue index, $[100\% - (MVIC_{post}/MVIC)] \times 100\%$; CK, creatine kinase. ^aPost lower than pre ($P < 0.05$), ^bPost greater than pre ($P < 0.05$), ^cMale post greater than female post ($P < 0.05$).

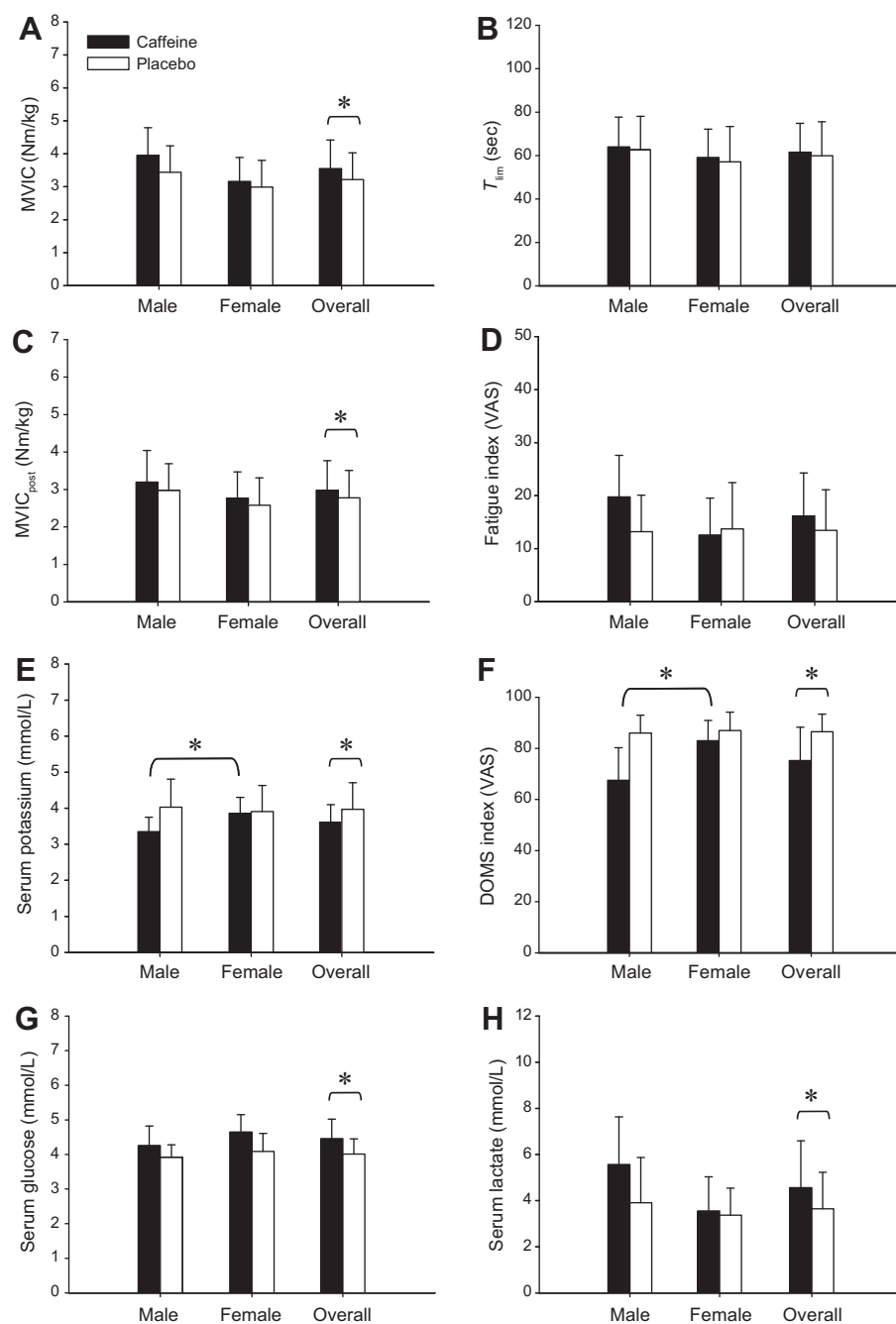


Fig. 2. Effects of sex and supplements on maximal voluntary isometric contractions (MVIC; A), submaximal voluntary isometric fatigue protocol (T_{lim} ; B), $MVIC_{post}$ (after T_{lim} and rest 20 s, do MVIC again; C), fatigue index ($[100\% - (MVIC_{post}/MVIC)] \times 100\%$; D), serum potassium (E), delayed-onset muscle soreness index (DOMS index; F), serum glucose (G), and serum lactate (H). Values are presented as means $\pm$ SD. *Significant difference ($P < 0.05$) by ANCOVA. All tests were performed under EIMD.

and $MVIC_{post}$ (7.2%) across both sexes, whereas there was no similar improvement found for T_{lim} (anaerobic endurance). Moreover, DOMS, when MVIC were performed, was reduced overall by 13% under the EIMD condition; furthermore, males experienced a much greater attenuating effect (21.5%) compared with females (4.6%). In addition, we have been able to demonstrate the presence of a reverse correlation between the attenuation of DOMS and restoration of MVIC, but for males only. Our result suggests that male athletes might benefit from the ergogenic effect of caffeine supplementation when affected by EIMD.

In the current study, after EIMD, we have shown that there is an increase in circulating CK concentrations (overall: 676.7 U/L) and DOMS (overall: 86.5), as well as a decrease in

muscle power during knee extensions as indexed by MVIC (overall: 22.6%; males: 21.0%; females: 23.9%) and $MVIC_{post}$ (overall: 22.5%; males: 21.9%; females: 23.0%), together with a deterioration in muscle endurance (i.e., T_{lim}) overall by 33.9% (males: 30.7%; females: 37%). Therefore, our downhill-running protocol was successfully able to induce muscle damage and a decrease in muscle performance.

In line with previous findings (31), this study has demonstrated that caffeine ingestion is able to facilitate a recovery in MVIC and $MVIC_{post}$ performance overall by 10.2 and 7.2%, respectively. This indicates that caffeine ingestion is an effective strategy for restoring impaired muscle power under the EIMD condition. The potential mechanisms that might be involved are that caffeine may enhance the activity of Na^+ -

K^+ -ATPase via a decrease in serum K^+ level, and/or it might inhibit the effect of adenosine on the CNS by blocking the perception of pain and fatigue, which is transmitted by adenosine receptors. The latter may consequently decrease the affected individual's perception of fatigue and DOMS, thus extending their exercise duration, facilitating glycolytic flux, and/or greater lactate accumulation during exercise (13). Our findings show a decrease in serum levels of K^+ after caffeine supplementation, as well as increased circulating glucose and lactate concentrations after exercise. These findings seem to serve as supporting evidence for the mechanisms that are proposed to underlie the improvements in MVIC and MVIC_{post} after caffeine supplementation. In terms of anaerobic muscle endurance performance, various studies have shown that caffeine has a beneficial effect on T_{lim} under non-EIMD conditions (6, 30, 32, 35). This is somewhat inconsistent with our findings. The discrepancy in muscle endurance performance under EIMD compared with non-EIMD may be attributed to the synergistical effects of muscle damage and DOMS combined with muscle fatigue induced by performing the three MVIC during our protocol before T_{lim} testing.

In terms of DOMS, in the current study, DOMS reached 86.5 while the MVIC was performed after placebo supplement; this can be attributed to the muscle inflammation associated with EIMD. We used a dose of 6 mg/kg of caffeine, and after such supplementation DOMS was decreased to 75.25 ($\downarrow$ 13%). This may be partly due to the fact that caffeine has a similar structure to adenosine and may be able to block the effect of adenosine on the CNS, which in turn would have an effect on the perception of pain propagated via the peripheral nerves to the CNS, the latter being transmitted by adenosine receptors (38). Importantly, the hormonal changes related to menstruation are likely to have an effect on the sensitivity to pain and the pain threshold of women (16). The current study, by controlling the potential contaminating effect of menstruation-related hormonal changes, has been able to reveal a sex difference regarding the attenuation of DOMS for knee extensions during the performance of MVIC after caffeine supplement, with male participants experiencing a much greater attenuation (21.5%) than their female counterparts (4.6%). It is noteworthy that the current study is among the first to demonstrate that male athletes experience a greater attenuation in DOMS than their female counterparts after caffeine ingestion under the EIMD condition. This could result from 1) female individuals having a greater sensitivity (20) and a lower tolerance to pain (46); and 2) caffeine, when acting as a competitive antagonist of adenosine receptors, is able to bring about higher locomotor activity in males compared with females (47). The aforementioned potential mechanisms are likely to be the reason there is an inverse correlation for males but not for females between attenuation of DOMS and the restoration of MVIC. This in turn implies that caffeine supplementation is likely to only have an ergogenic effect on male athletes under EIMD. In addition, in the present study, the greater decrease in serum K^+ levels in the male participants could be another potential explanation. A reduction in the circulating K^+ concentration could affect adenosine in the CNS and block the perception of pain and fatigue transmitted by adenosine receptors (13) during recovery exercise under EIMD. In line with our findings, other studies have also reported a negative association between DOMS and muscle performance (15). Inter-

estingly, one study using a similar exercise testing protocol as ours, and after recruiting only female individuals, demonstrated that a caffeine supplement with a dosage of 5 mg/kg was able to attenuate DOMS for knee extensions when MVIC were performed by 47.7% (31), which is a ratio much greater than ours. This discrepancy between Maridakis et al. (31) and the current study can be attributed to the differences in methodologies used to induce muscle damage, which resulted in much lower DOMS indices following EIMD in the Maridakis et al. study (26.6) than the current study (86.5).

It is also worth noting that our study demonstrated greater muscle performance recovery in males than females for MVIC (males: 14.8%; females: 5.7%), albeit the fact that no statistical significant differences were found. Neuropharmacological and psychopharmacological studies have indicated that sex may play a moderating role in the relationship between caffeine supplement and arousal change in that males seem to perceive reduced somnolence and greater activation 10–30 min after supplementation compared with females (1). Indeed, we observed that there was greater muscle performance recovery in males than females for MVIC (but also lacking statistical significance). To the best of our knowledge, only one study has recruited untrained females to investigate the effect of caffeine supplementation on anaerobic muscle performance under EIMD conditions; this was able to demonstrate improvement in MVIC for knee extensions of only 4.4% (31). The results are similar to our findings for our female participants (MVIC: 5.7% under EIMD), implying that female individuals might potentially experience a lower ergogenic benefit from acute caffeine ingestion under EIMD. Nonetheless and interestingly, no sex difference was observed regarding recovery of MVIC_{post} performance (male: 7.4%; female: 7.4%). The precise reasons for this remain unclear, and a future study might be needed to further investigate this topic.

The circulating concentration of CK has been used as a marker for inflammation and muscle damage after the onset of EIMD in a number of studies (11, 12). A sex difference has been found in serum levels of CK, with females manifesting a smaller increase ($\uparrow$ 221%) than males ($\uparrow$ 513%), which indicates that female athletes experienced muscle inflammation and damage to a lesser extent. This finding may be attributed to females having a higher concentration of estrogen, a hormone that has antioxidative and membrane-stabilizing properties (27); this may help to attenuate muscle damage, reduce myofiber destruction, and lower inflammation, as well as facilitate the repair of tissue damage (44). Another explanation that can be put forward for this difference is that the female participants who were recruited into the current study had lower body weights, and this might have resulted in less damage to their bodies during the downhill running. The current study also observed a relatively high DOMS value (male: 82.0 and female: 85.8) when MVIC were performed, and this could indicate the possibility of induced acute inflammation caused by the lower limb muscle damage; notwithstanding this possibility, the decreases in anaerobic muscle power and DOMS were similar between sexes.

There are some limitations due to the unavoidable physiological responses which might occur in the caffeine-based investigation (i.e., increase in heart rate after supplementation) (37), and some participants could potentially (e.g., psychologically) be affected during the trials. To avoid the above-

mentioned associated side effects for muscle performance and to increase the reliability of the results, verbal encouragements were equally given throughout the exercise tests in our study. Studies have shown that the peak level of DOMS will be between 24 and 48 h after EIMD (8, 9). Considering the potential differences in the physiological response to the caffeine supplementation (i.e., digestion and faster and slower metabolizers), the within-participant design was therefore employed in the present study. Participants can serve as their own control (caffeine vs. placebo) and avoid the repeated-bout effect from inducing DOMS twice. However, we still could not discount that the level of DOMS could potentially be different at 24 and 48 h when prescribing the supplements. Moreover, future studies could consider adding an electronic algometry test (10), alongside a visually subjective scale for more objectively assessing DOMS (i.e., pressure pain threshold).

Conclusion

Acute caffeine supplementation at a dosage of 6 mg/kg is able to facilitate recovery of anaerobic power (MVIC, MVIC_{post}) and attenuate DOMS when MVIC are performed; by way of contrast, there was no effect manifested on anaerobic endurance (i.e., T_{lim}). Furthermore, male athletes, compared with female athletes, experienced a greater reduction in DOMS for enhanced MVIC when caffeine was prescribed. This suggests that male athletes might benefit from the ergogenic effect of acute caffeine supplementation after the onset of EIMD.

ACKNOWLEDGMENTS

We thank all the participants for the time and effort while taking part in this project. In addition, we appreciate the support from the exercise physiology and exercise biomechanics laboratories at National Taiwan Normal University.

GRANTS

This project was funded by the Ministry of Science and Technology TAIWAN (MOST)-NSC101-2410-H-003-141.

DISCLOSURES

No conflicts of interest, financial or otherwise, are declared by the author(s).

AUTHOR CONTRIBUTIONS

H.-Y.C. and H.-S.W. conceived and designed research; H.-Y.C., K.T., and H.-H.C. performed experiments; H.-Y.C. analyzed data; H.-Y.C., Y.-C.C., and H.-S.W. interpreted results of experiments; H.-Y.C. prepared figures; H.-Y.C. and H.-S.W. drafted manuscript; H.-Y.C., Y.-C.C., and H.-S.W. edited and revised manuscript; H.-Y.C., Y.-C.C., K.T., H.-H.C. and H.-S.W. approved final version of manuscript.

REFERENCES

- Adan A, Prat G, Fabbri M, Sánchez-Turet M. Early effects of caffeinated and decaffeinated coffee on subjective state and gender differences. *Prog Neuropsychopharmacol Biol Psychiatry* 32: 1698–1703, 2008. doi:10.1016/j.pnpbp.2008.07.005.
- Archana S, Jaspal SS. Effects of caffeine ingestion on strength and endurance performance of normal young adults. *Doping J* 7: 2, 2010.
- Boyas S, Guével A. Influence of exercise intensity and joint angle on endurance time prediction of sustained submaximal isometric knee extensions. *Eur J Appl Physiol* 111: 1187–1196, 2011. doi:10.1007/s00421-010-1731-0.
- Byrne C, Eston R. The effect of exercise-induced muscle damage on isometric and dynamic knee extensor strength and vertical jump performance. *J Sports Sci* 20: 417–425, 2002. doi:10.1080/026404102317366672.
- Burke LM. Caffeine and sports performance. *Appl Physiol Nutr Metab* 33: 1319–1334, 2008. doi:10.1139/H08-130.
- Chen HY, Wang HS, Tung K, Chao HH. Effects of gender difference and caffeine supplementation on anaerobic muscle performance. *Int J Sports Med* 36: 974–978, 2015. doi:10.1055/s-0035-1550048.
- Chen TC, Lin KY, Chen HL, Lin MJ, Nosaka K. Comparison in eccentric exercise-induced muscle damage among four limb muscles. *Eur J Appl Physiol* 111: 211–223, 2011. doi:10.1007/s00421-010-1648-7.
- Chen TC, Nosaka K, Tu JH. Changes in running economy following downhill running. *J Sports Sci* 25: 55–63, 2007. doi:10.1080/02640410600718228.
- Chen TC, Nosaka K, Wu CC. Effects of a 30-min running performed daily after downhill running on recovery of muscle function and running economy. *J Sci Med Sport* 11: 271–279, 2008. doi:10.1016/j.jsams.2007.02.015.
- Chesterton LS, Sim J, Wright CC, Foster NE. Interrater reliability of algometry in measuring pressure pain thresholds in healthy humans, using multiple raters. *Clin J Pain* 23: 760–766, 2007. doi:10.1097/AJP.0b013e318154b6ae.
- Cheung K, Hume P, Maxwell L. Delayed onset muscle soreness: treatment strategies and performance factors. *Sports Med* 33: 145–164, 2003. doi:10.2165/00007256-200333020-00005.
- Connolly DA, Sayers SP, McHugh MP. Treatment and prevention of delayed onset muscle soreness. *J Strength Cond Res* 17: 197–208, 2003.
- Davis JK, Green JM. Caffeine and anaerobic performance: ergogenic value and mechanisms of action. *Sports Med* 39: 813–832, 2009. doi:10.2165/11317770-000000000-00000.
- Essig D, Costill DL, Van-Handel PJ. Effects of caffeine ingestion on utilization of muscle glycogen and lipid during leg ergometer cycling. *Int J Sports Med* 01: 86–90, 1980. doi:10.1055/s-2008-1034637.
- Farina D, Arendt-Nielsen L, Merletti R, Graven-Nielsen T. Effect of experimental muscle pain on motor unit firing rate and conduction velocity. *J Neurophysiol* 91: 1250–1259, 2004. doi:10.1152/jn.00620.2003.
- Fillingim RB, King CD, Ribeiro-Dasilva MC, Rahim-Williams B, Riley JL III. Sex, gender, and pain: a review of recent clinical and experimental findings. *J Pain* 10: 447–485, 2009. doi:10.1016/j.jpain.2008.12.001.
- Ganio MS, Johnson EC, Lopez RM, Stearns RL, Emmanuel H, Anderson JM, Casa DJ, Maresh CM, Volek JS, Armstrong LE. Caffeine lowers muscle pain during exercise in hot but not cool environments. *Physiol Behav* 102: 429–435, 2011. doi:10.1016/j.physbeh.2010.12.005.
- Gliottoni RC, Meyers JR, Arngrimsson SA, Broglio SP, Motl RW. Effect of caffeine on quadriceps muscle pain during acute cycling exercise in low versus high caffeine consumers. *Int J Sport Nutr Exerc Metab* 19: 150–161, 2009. doi:10.1123/ijnsnm.19.2.150.
- Gliottoni RC, Motl RW. Effect of caffeine on leg-muscle pain during intense cycling exercise: possible role of anxiety sensitivity. *Int J Sport Nutr Exerc Metab* 18: 103–115, 2008. doi:10.1123/ijnsnm.18.2.103.
- Greenspan JD, Craft RM, LeResche L, Arendt-Nielsen L, Berkley KJ, Fillingim RB, Gold MS, Holdcroft A, Lautenbacher S, Mayer EA, Mogil JS, Murphy AZ, Traub RJ; Consensus Working Group of the Sex, Gender, and Pain SIG of the IASP. Studying sex and gender differences in pain and analgesia: a consensus report. *Pain* 132, Suppl 1: S26–S45, 2007. doi:10.1016/j.pain.2007.10.014.
- Grgic J, Mikulic P, Schoenfeld BJ, Bishop DJ, Pedisic Z. The influence of caffeine supplementation on resistance exercise: a review. *Sports Med* 49: 17–30, 2019. doi:10.1007/s40279-018-0997-y.
- Grgic J, Pickering C. The effects of caffeine ingestion on isokinetic muscular strength: a meta-analysis. *J Sci Med Sport* 22: 353–360, 2019. doi:10.1016/j.jsams.2018.08.016.
- Grgic J, Trexler ET, Lazinica B, Pedisic Z. Effects of caffeine intake on muscle strength and power: a systematic review and meta-analysis. *J Int Soc Sports Nutr* 15: 11, 2018. doi:10.1186/s12970-018-0216-0.
- Hicks KM, Onambélé GL, Winwood K, Morse CI. Muscle damage following maximal eccentric knee extensions in males and females. *PLoS One* 11: e0150848, 2016. doi:10.1371/journal.pone.0150848.
- Hurley CF, Hatfield DL, Riebe DA. The effect of caffeine ingestion on delayed onset muscle soreness. *J Strength Cond Res* 27: 3101–3109, 2013. doi:10.1519/JSC.0b013e3182a99477.
- Karabalaefar S, Hefzollasan M, Behpoor N, Ghaleghir S. Effect of caffeine on the amount of perceived pain, joint range of motion and edema after delayed muscle soreness. *Pedagogics Psychol Med Biol Problems Physical Training Sports* 1: 96–100, 2013.
- Kendall B, Eston R. Exercise-induced muscle damage and the potential protective role of estrogen. *Sports Med* 32: 103–123, 2002. doi:10.2165/00007256-200232020-00003.

28. Lamb GD, Cellini MA, Stephenson DG. Different Ca^{2+} releasing action of caffeine and depolarisation in skeletal muscle fibres of the rat. *J Physiol* 531: 715–728, 2001. doi:[10.1111/j.1469-7793.2001.0715h.x](https://doi.org/10.1111/j.1469-7793.2001.0715h.x).
29. Lännergren J, Bruton JD, Westerblad H. Vacuole formation in fatigued skeletal muscle fibres from frog and mouse: effects of extracellular lactate. *J Physiol* 526: 597–611, 2000. doi:[10.1111/j.1469-7793.2000.00597.x](https://doi.org/10.1111/j.1469-7793.2000.00597.x).
30. Madigan DJ, Willems MET. Effect of caffeine on fatigue during submaximal isometric contractions at different knee angles. *Med Sport (Roma)* 15: 194–200, 2011. doi:[10.2478/v10036-011-0027-8](https://doi.org/10.2478/v10036-011-0027-8).
31. Maridakis V, O'Connor PJ, Dudley GA, McCully KK. Caffeine attenuates delayed-onset muscle pain and force loss following eccentric exercise. *J Pain* 8: 237–243, 2007. doi:[10.1016/j.jpain.2006.08.006](https://doi.org/10.1016/j.jpain.2006.08.006).
32. Meyers BM, Cafarelli E. Caffeine increases time to fatigue by maintaining force and not by altering firing rates during submaximal isometric contractions. *J Appl Physiol* (1985) 99: 1056–1063, 2005. doi:[10.1152/japplphysiol.00937.2004](https://doi.org/10.1152/japplphysiol.00937.2004).
33. Miller C, Newton SE. Pain perception and expression: the influence of gender, personal self-efficacy, and lifespan socialization. *Pain Manag Nurs* 7: 148–152, 2006. doi:[10.1016/j.pmn.2006.09.004](https://doi.org/10.1016/j.pmn.2006.09.004).
34. Murphy RM, Dutka TL, Horvath D, Bell JR, Delbridge LM, Lamb GD. Ca^{2+} -dependent proteolysis of junctophilin-1 and junctophilin-2 in skeletal and cardiac muscle. *J Physiol* 591: 719–729, 2013. doi:[10.1113/jphysiol.2012.243279](https://doi.org/10.1113/jphysiol.2012.243279).
35. Plaskett CJ, Cafarelli E. Caffeine increases endurance and attenuates force sensation during submaximal isometric contractions. *J Appl Physiol* (1985) 91: 1535–1544, 2001. doi:[10.1152/jappl.2001.91.4.1535](https://doi.org/10.1152/jappl.2001.91.4.1535).
36. Sabblah S, Dixon D, Bottoms L. Sex differences on the acute effects of caffeine on maximal strength and muscular endurance. *Comp Exerc Physiol* 11: 89–94, 2015. doi:[10.3920/CEP150010](https://doi.org/10.3920/CEP150010).
37. Saunders B, de Oliveira LF, da Silva RP, de Salles Painelli V, Gonçalves LS, Yamaguchi G, Mutti T, Maciel E, Roschel H, Artioli GG, Gualano B. Placebo in sports nutrition: a proof-of-principle study involving caffeine supplementation. *Scand J Med Sci Sports* 27: 1240–1247, 2017. doi:[10.1111/sms.12793](https://doi.org/10.1111/sms.12793).
38. Sawynok J. Caffeine and pain. *Pain* 152: 726–729, 2011. doi:[10.1016/j.pain.2010.10.011](https://doi.org/10.1016/j.pain.2010.10.011).
39. Sökmen B, Armstrong LE, Kraemer WJ, Casa DJ, Dias JC, Judelson DA, Maresh CM. Caffeine use in sports: considerations for the athlete. *J Strength Cond Res* 22: 978–986, 2008. doi:[10.1519/JSC.0b013e3181660cec](https://doi.org/10.1519/JSC.0b013e3181660cec).
40. Spriet LL, MacLean DA, Dyck DJ, Hultman E, Cederblad G, Graham TE. Caffeine ingestion and muscle metabolism during prolonged exercise in humans. *Am J Physiol Endocrinol Metab Physiol* 262: E891–E898, 1992. doi:[10.1152/ajpendo.1992.262.6.E891](https://doi.org/10.1152/ajpendo.1992.262.6.E891).
41. Stephenson DG. Caffeine—a valuable tool in excitation-contraction coupling research. *J Physiol* 586: 695–696, 2008. doi:[10.1113/jphysiol.2007.148270](https://doi.org/10.1113/jphysiol.2007.148270).
42. Tarnopolsky M, Cupido C. Caffeine potentiates low frequency skeletal muscle force in habitual and nonhabitual caffeine consumers. *J Appl Physiol* (1985) 89: 1719–1724, 2000. doi:[10.1152/jappl.2000.89.5.1719](https://doi.org/10.1152/jappl.2000.89.5.1719).
43. Tallis J, Duncan MJ, James RS. What can isolated skeletal muscle experiments tell us about the effects of caffeine on exercise performance? *Br J Pharmacol* 172: 3703–3713, 2015. doi:[10.1111/bph.13187](https://doi.org/10.1111/bph.13187).
44. Tiidus PM. Can oestrogen influence skeletal muscle damage, inflammation, and repair? *Br J Sports Med* 39: 251–253, 2005. doi:[10.1136/bjsm.2005.016881](https://doi.org/10.1136/bjsm.2005.016881).
45. Warren GL, Park ND, Maresca RD, McKibans KI, Millard-Stafford ML. Effect of caffeine ingestion on muscular strength and endurance: a meta-analysis. *Med Sci Sports Exerc* 42: 1375–1387, 2010. doi:[10.1249/MSS.0b013e3181cabb8d](https://doi.org/10.1249/MSS.0b013e3181cabb8d).
46. Wiesenfeld-Hallin Z. Sex differences in pain perception. *Gend Med* 2: 137–145, 2005. doi:[10.1016/S1550-8579\(05\)80042-7](https://doi.org/10.1016/S1550-8579(05)80042-7).
47. Yang JN, Chen JF, Fredholm BB. Physiological roles of A_1 and $\text{A}_{2\text{A}}$ adenosine receptors in regulating heart rate, body temperature, and locomotion as revealed using knockout mice and caffeine. *Am J Physiol Heart Circ Physiol* 296: H1141–H1149, 2009. doi:[10.1152/ajpheart.00754.2008](https://doi.org/10.1152/ajpheart.00754.2008).